

A primer on structured decision making for invasive species (using dynamic programming)

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Invasive species (IAS) pose significant ecological, economic, and public health challenges.

Robust methods are needed for guiding effective management. Overall, I hope to demonstrate the flexibility and utility of structured decision making in invasive species management.

Target journal: soit NeoBiota <https://www.reabic.net/journals/mbi/Submission.aspx> (1200 euros, plusieurs formats d'article pertinents) ou encore Biological invasions au format INVASION NOTES, ou encore Peer Community journal.

voir NeoBiota, ; repartir de cet article using structured decision-making to gain consensus on approaches to invasive species control, des articles qu'il cite, et des articles qui l'ont cité ; pour PC Journal ? Faire un petit shiny qui va bien?

Keywords: Invasive species, Structured decision making, Adaptive management, Stochastic dynamic programming

Introduction

Invasive species are a significant global issue, with wide-ranging impacts on ecosystems, economies, and public health (Pyšek et al. 2020, Roy et al. 2024).

Effective management of invasive species requires the estimation of population abundance for guiding control efforts and evaluating the success of eradication or regulation programs (Williams et al. 2002, Thompson et al. 2021).

Pitch: SDM/AM plusieurs fois recommandées pour management of invasive species. Mais encore peu utilisée. Many reasons. Une d'entre elles est que c'est technique la partie determine trade-offs. Here on participe à l'effort en illustrant la SDP et qqs-unes des variantes (active/passive, POMDP) et dans des cas différents (single pop, metapop, two interacting pops). Plusieurs papiers qui préconisent/ont fait de la SDM sur invasive species. Faire une petite review.

A primer on structured decision making

Why and what

Stochastic dynamic programming

Example on computer maintenance pour bien expliquer la SDP

Case studies with coyotes

Single population

Reprendre l'exemple loup de Lulu dans l'article SDP

36 **Two interacting species**

37 Competition or predator-prey, see SDP-species-interactions/ dans la biblio

38 **Metapopulation**

39 **Accounting for uncertainty**

40 **POMDP**

41 Illustrer POMDP (<https://chatgpt.com/share/68814183-9078-8011-9697-c3ee73b27762>)

42 **Adaptive management**

43 Sur une espèce (voir <https://chatgpt.com/share/6873c9fd-4df4-8011-97cb-2b6147d505fe> ; voir
44 aussi le bouquin de Conroy sur SDM, il y a une figure qui l'explique bien).

45 **Discussion**

46 **Main messages**

47 **Limitations**

48 **Perspectives**

49 **Acknowledgments**

50 **Data availability statement**

51 Data and code are available at <https://github.com/oliviergimenez/counting-rodents>.

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